Netflix Content Strategy Analytics

Data Visualisation — Project 2, Phase 3 Report

Team 5

March 2026

Website: https://YOUR-HOSTED-LINK-HERE.com
YouTube Demo (Unlisted): https://youtu.be/YOUR-VIDEO-ID
How to run locally:

```
cd Phase2/
python3 -m http.server 8000
# Open http://localhost:8000/Phase2/ in a browser
```

1. Dataset

We use two Kaggle CSV files covering Netflix's catalog up to 2025:

- **netflix_movies_detailed_up_to_2025.csv** — ∼16 000 movies
- **netflix_tv_shows_detailed_up_to_2025.csv** — ∼16 000 TV shows

Key fields: `title`, `date_added`, `release_year`, `vote_average` (IMDb-style rating), `vote_count`, `genres` (multi-valued), `country` (multi-valued), `language`, `popularity`, `budget`, `revenue`.
Preprocessing: We parse `date_added` into year and month, split multi-valued `genres` and `country` fields, filter to 2010–2025, and compute derived metrics (Simpson's Diversity Index, quantile statistics). Titles with missing or invalid years are excluded (<1% of rows).

2. Visualisations

Our dashboard contains four interactive D3.js (v7) visualisations, each answering a distinct analytical question.

Chart 1: Content Quality & Volume (Bubble Chart)

Question: *How has the volume and quality of Netflix content evolved over time?*

Each bubble represents a cohort of titles grouped by release year (x-axis) and rounded IMDb rating (y-axis, 0.5-step bins). Bubble **area** encodes the number of titles (using a $\sqrt{}$-scale so area $\propto$ count, avoiding the perceptual distortion of linear radius mapping). Users toggle between **Both** (default), Movies (coral-red), and TV Shows (steel-blue). A semi-transparent green band highlights the "high quality" zone ($\geq$7.0).

Real-world use: Content strategists can identify whether Netflix's recent acquisitions maintain quality standards or if volume growth comes at the cost of lower-rated titles.

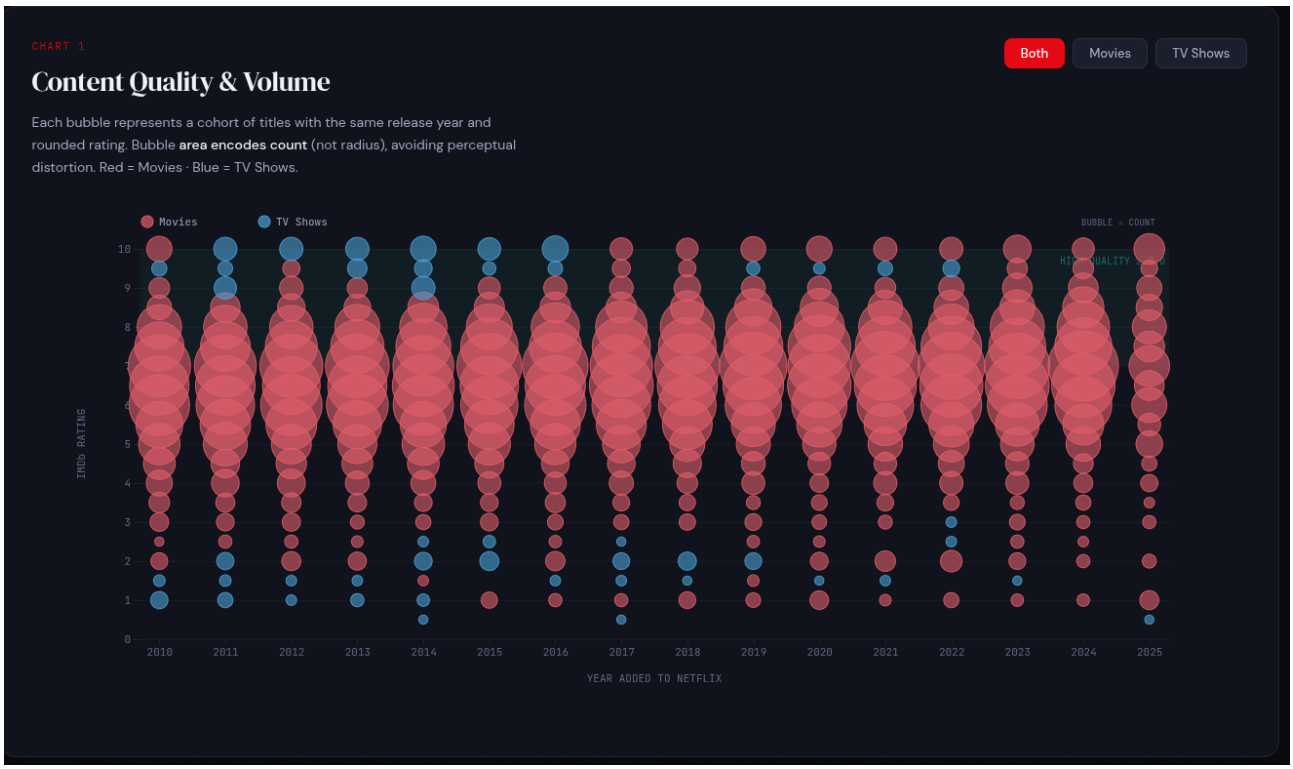


Chart 2: Genre Ecosystem & Balance (Sunburst / Treemap)

Question: *How is Netflix's catalog distributed across genres, and what is the Movies-vs-Shows split within each genre?*

A hierarchical visualisation with two modes: **Sunburst** (preserves parent-child hierarchy via angular encoding) and **Treemap** (optimises area comparison using squarified tiling with golden-ratio aspect). The top 15 genres are shown; remaining genres collapse into "Other." Clicking a genre segment zooms in to show its Movies/TV Shows breakdown.

Real-world use: Helps catalogue managers identify genre gaps and over-served categories for informed acquisition decisions.

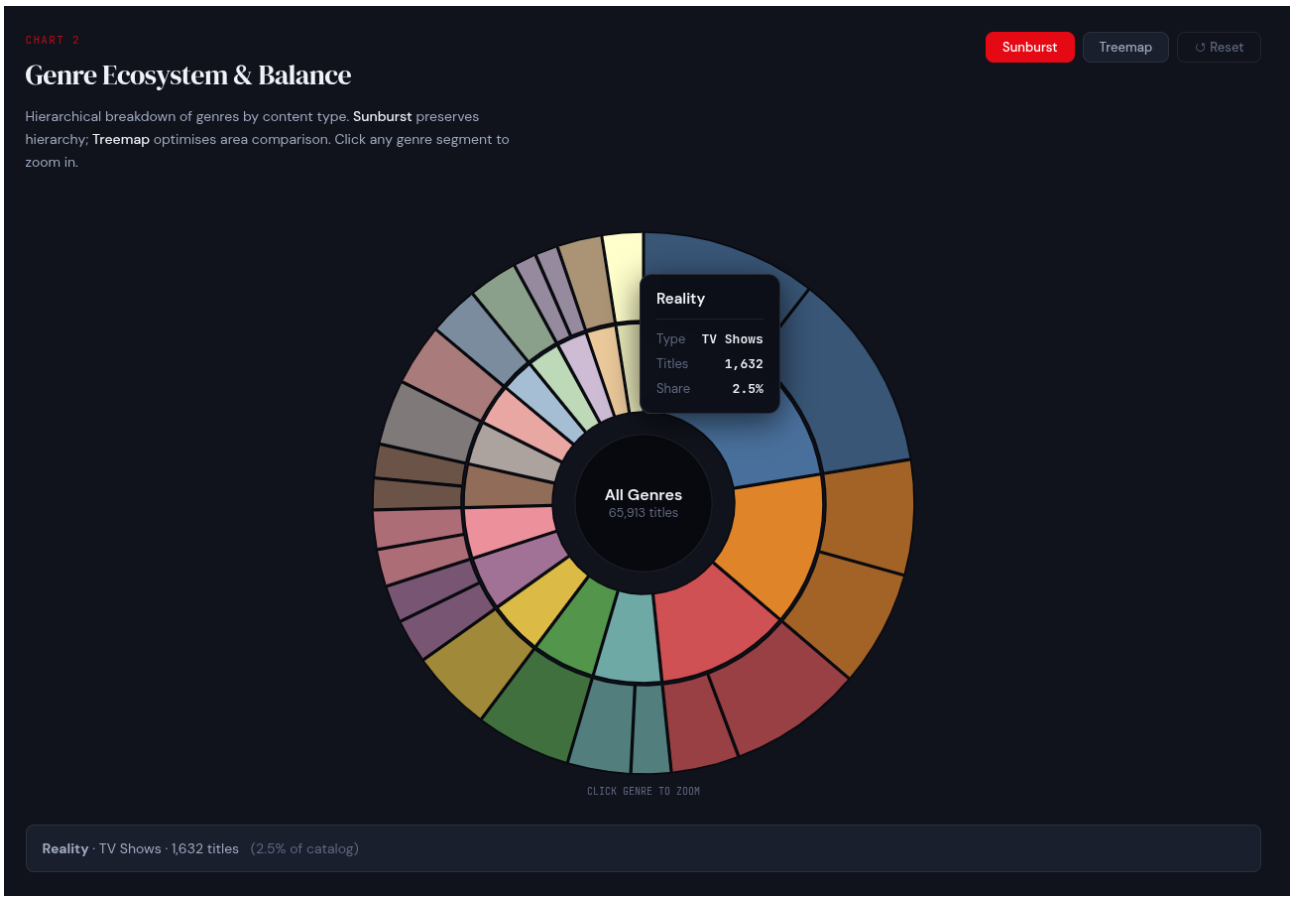


Chart 3: Geographic Diversity & Market Evolution (Streamgraph)

Question: *How has the geographic diversity of Netflix's content evolved? Is Netflix becoming more internationally diverse?*

A streamgraph (wiggle offset) or stacked area chart showing the top 8 production countries over time. An optional **Simpson's Diversity Index** overlay (dashed amber line on a secondary axis) quantifies concentration: values approaching 1.0 indicate a more uniform international distribution. Hovering a country layer highlights it and dims others for clarity.

Real-world use: International expansion teams can track whether Netflix is achieving its stated goal of producing more local-language content worldwide.

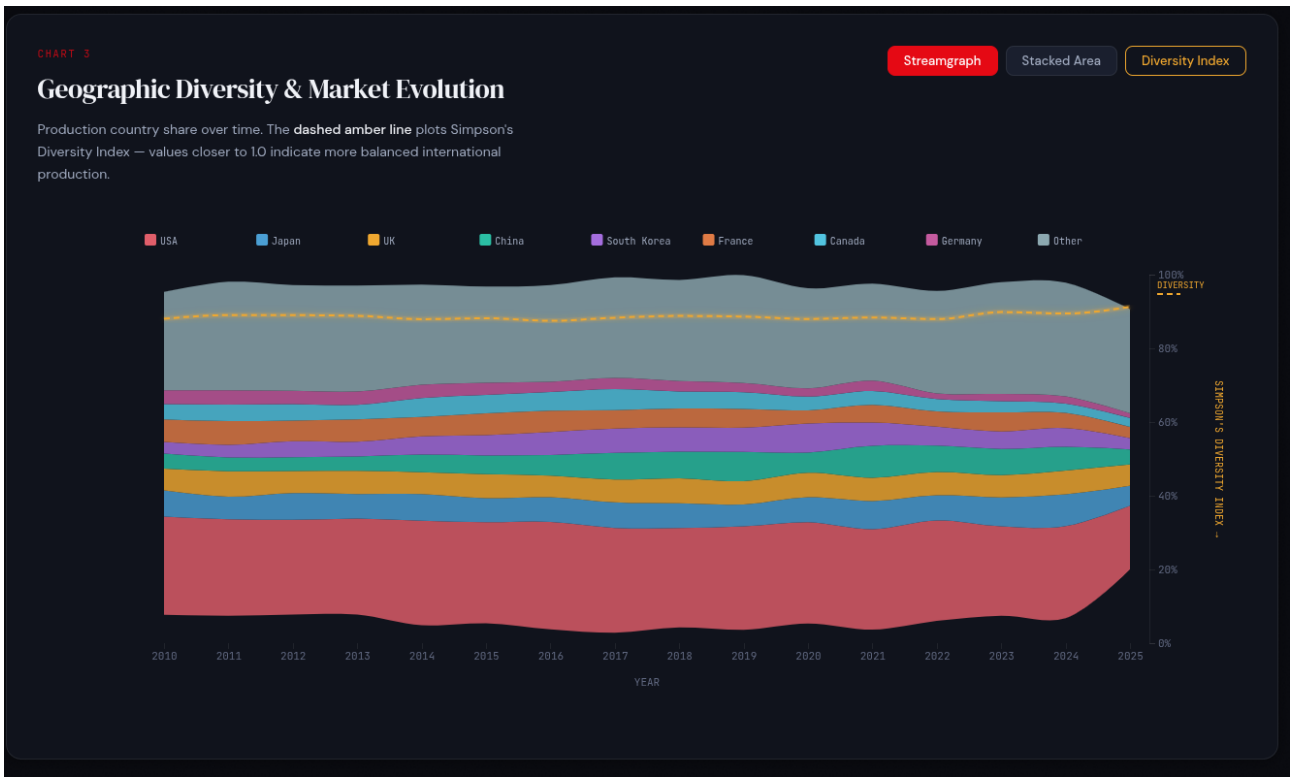
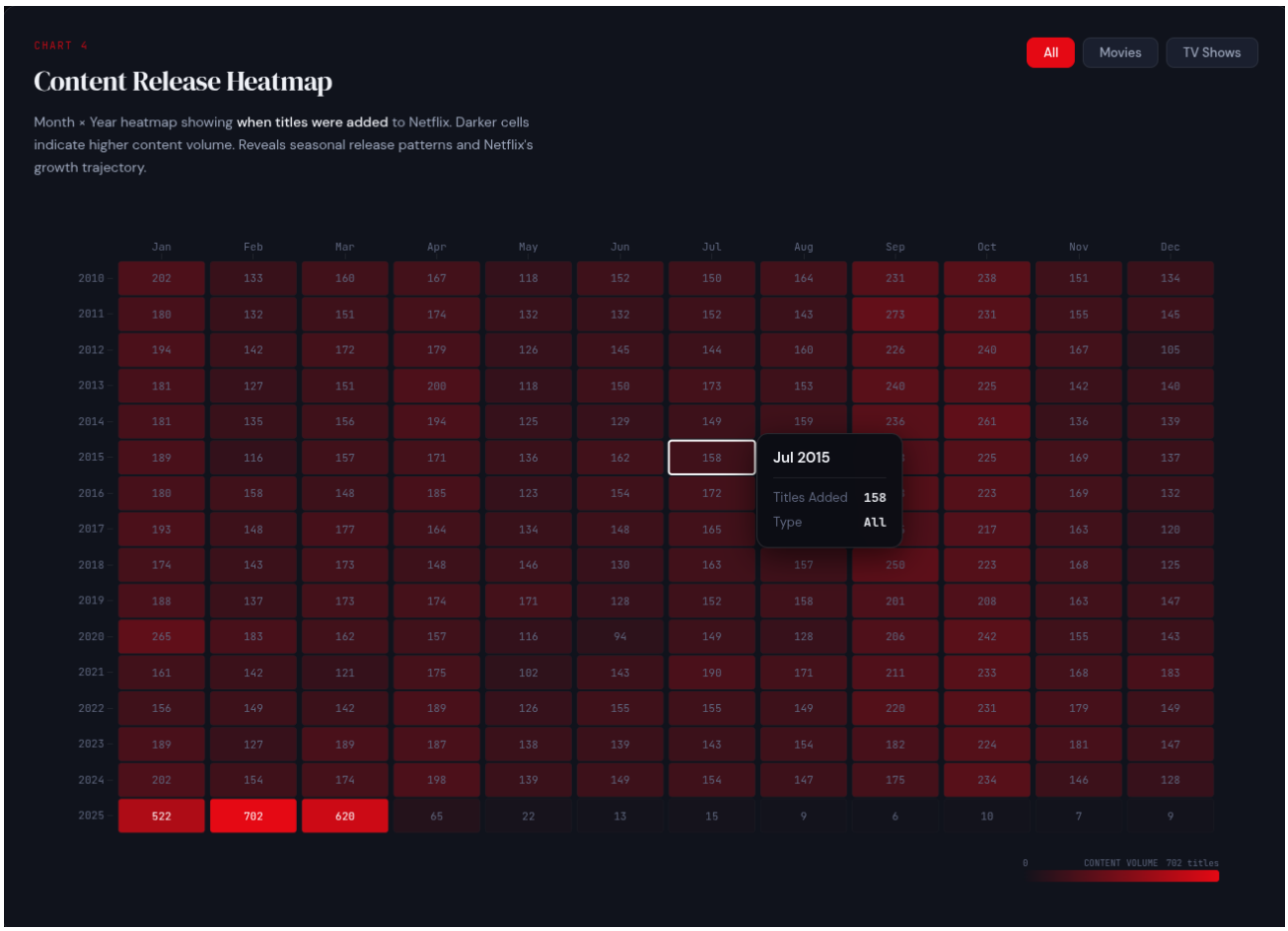


Chart 4: Content Release Heatmap (Month × Year)

Question: *When does Netflix add the most content? Are there seasonal release patterns?*

A month-by-year heatmap using a sequential colour scale (dark background → Netflix red). Each cell shows the count of titles added in that month/year. Users can filter by Movies, TV Shows, or All. Darker cells reveal peak acquisition months.

Real-world use: Marketing teams can align promotional campaigns with content drops; competitors can study Netflix's release cadence.



Cross-Filtering (Linked Views)

All four widgets share a common `crossFilter` state. Clicking a **genre segment** in Chart 2 (sunburst or treemap) filters Charts 1, 3, 4 and the KPI header to show only titles in that genre. Similarly, clicking a **country layer** in Chart 3 filters all other charts. A persistent status bar (`#crossfilter-bar`) shows active filters and provides a "Clear All" button. Re-clicking the same element toggles the filter off. This architecture enables rapid drill-down analysis without losing context.

3. Design Choices & Justifications

Principle	Where Applied	Rationale
Area encoding (Stevens' power law)	Chart 1 bubble sizing	$\sqrt{}$-scale ensures perceived area $\propto$ count. Linear radius would make large cohorts appear $r^2\times$ larger.
Gestalt — Proximity	KPI strip, grouped legend items	Metrics grouped into a single strip signal they describe the same dataset.
Gestalt — Similarity	Consistent colour tokens across charts	Movies are always coral-red, TV Shows always steel-blue — reduces cognitive load.
Colorblind safety	Categorical palette avoids red/green	Coral-red vs. steel-blue is distinguishable under protanopia and deuteranopia.
Hierarchy via encoding	Sunburst angular + radial encoding	Angular extent encodes proportion; radial depth encodes hierarchy level (Munzner Ch. 7).
Squarified treemap	Chart 2 treemap mode	Golden-ratio aspect ($\phi \approx 1.618$) maximises cell readability (Bruls et al.).
Pre-attentive features	Heatmap sequential colour	Intensity (lightness) is pre-attentively processed, making high-count cells "pop."
Tooltip — fixed position	All charts	Uses `clientX/Y` (viewport-relative) instead of `pageX/Y` to prevent scroll-offset drift.
Data-ink ratio	Minimal axes, no chart borders	Following Tufte's principle: grid lines at $\alpha = 0.04$ provide reference without clutter.
Accessibility	`aria-label`, `aria-pressed`	SVG elements carry `role="img"` + descriptive labels; toggle buttons expose state to screen readers.

4. Challenges & Limitations

Challenges

1. **Multi-valued fields:** Genres and countries are comma-separated strings. A single title can appear in multiple genre/country counts, requiring careful deduplication and clear communication that totals reflect "title-genre pairs," not unique titles.
2. **Performance with 32 000+ rows:** Rendering jittered points and large heatmaps required sub-sampling and efficient D3 `join()` patterns to maintain 60 fps interactions.
3. **Perceptual encoding:** Ensuring bubble area, not radius, encodes count required a $\sqrt{}$-scale — an easy mistake that inflates visual perception of large values.
4. **Responsive SVGs:** Charts must redraw on window resize. We use a debounced resize listener and percentage-based widths to handle this.
5. **Cross-origin data loading:** Browsers block `file://` fetch requests; the project must be served via HTTP (e.g., `python3 -m http.server`).

5. Team Contributions

Member	Contribution
Vineel	Data preprocessing, Chart 1 (Bubble Chart), Chart 4 (Heatmap)
Babu	Chart 2 (Sunburst/Treemap), Chart 3 (Streamgraph), Diversity Index
Jithendra	UI/UX Design, CSS, Website Hosting, Video Presentation